

Polishchuk

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1	2	3	Σ
7	3,5	5	15,5

sehr gut!

Aufgabe 1.1

Rodrigues - Formel: $P_n(x) := \frac{1}{2^n \cdot n!} \cdot \frac{d^n}{dx^n} (x^2 - 1)^n$

$$P_0(x) = 1$$

$$P_1(x) = x$$

$$P_2(x) = \frac{1}{2} (3x^2 - 1)$$

Ziel: zz. dass P_n ein Orthogonalsystem in $\mathbb{R}[-1, 1]$ bildet:

$$\int_{-1}^1 P_n(x) \cdot P_m(x) dx = 0 \quad \text{für } n \neq m$$

a). Gegeben: $\int_{-1}^1 \frac{d^n}{dx^n} (x^2 - 1) \cdot \frac{d^m}{dx^m} (x^2 - 1)^m dx$

$$= (-1)^k \int_{-1}^1 \frac{d^{n-k}}{dx^{n-k}} (x^2 - 1)^n \cdot \frac{d^{m+k}}{dx^{m+k}} (x^2 - 1)^m dx$$

$$\forall n, m \in \mathbb{N}_0, \quad n \geq m, \quad 0 \leq k \leq n.$$

Beweis: Wählen $k=n$: $\frac{d^{n-k}}{dx^{n-k}} (x^2 - 1)^n = \frac{d^0}{dx^0} (x^2 - 1)^n = (x^2 - 1)^n$

Bei $x = \pm 1$ $(x^2 - 1)^n$ hat die Nullstelle.

$(x^2 - 1)^m$ ist ein Polynom Grad $2m$. Da $n \geq m$, gilt $m + n \geq 2m$ ✓

$$\begin{aligned} \int_{-1}^1 P_n(x) \cdot P_m(x) dx &= \int_{-1}^1 \frac{1}{2^n \cdot n!} \cdot \frac{d^n}{dx^n} (x^2 - 1)^n \cdot \frac{1}{2^m m!} \cdot \frac{d^m}{dx^m} (x^2 - 1)^m dx = \\ &= \frac{1}{2^n n!} \cdot \frac{1}{2^m m!} \cdot \int_{-1}^1 \frac{d^n}{dx^n} (x^2 - 1)^n \cdot \frac{d^m}{dx^m} dx \\ &= \frac{1}{2^n n!} \cdot \frac{1}{2^m m!} \cdot (-1)^k \int_{-1}^1 \frac{d^{n-k}}{dx^{n-k}} (x^2 - 1)^n \cdot \frac{d^{m+k}}{dx^{m+k}} (x^2 - 1)^m dx \end{aligned}$$

Da $n \geq m$ und im Integral rechnen wir für $m \neq n$ gilt $n > m$
 $\therefore n + m > 2m$:

$$\frac{d^{m+n}}{dx^{m+n}} (x^2 - 1) = 0 \quad \checkmark$$

$$\therefore \int_{-1}^1 P_n(x) P_m(x) dx =$$

$$= \frac{1}{2^n n!} \cdot \frac{1}{2^m m!} \cdot (-1)^n \cdot \int_{-1}^1 (x^2 - 1)^n \cdot 0 dx = 0 \quad \checkmark \quad \square$$

b). 22. für fest. $n \in \mathbb{N}_0$ mittels "endl. F." dass:

$$\int_{-1}^1 (1-x)^n (1+x)^n dx = \frac{(n!)^2}{(n-k)!(n+k)!} \cdot \int_{-1}^1 (1-x)^{n-k} (1+x)^{n+k} dx, \text{ für } k=0, \dots, n$$

$$\text{IA: } k=0: \quad \text{RHS: } \frac{(n!)^2}{(n!) \cdot (n!)} \cdot \int_{-1}^1 (1-x)^n \cdot (1+x)^n dx$$

$$= \frac{1}{2} \cdot \int_{-1}^1 (1-x)^n (1+x)^n dx = \text{LHS} \quad \checkmark$$

IV: Sei gilt für k :

$$\int_{-1}^1 (1-x)^n (1+x)^n dx = \frac{(n!)^2}{(n-k)!(n+k)!} \cdot \int_{-1}^1 (1-x)^{n-k} (1+x)^{n+k} dx$$

IS: $k+1$: RHS: $\frac{(n!)^2}{(n-k-1)!(n+k+1)!} \cdot \int_{-1}^1 (1-x)^{n-k-1} \cdot (1+x)^{n+k+1} dx$

Sei $u = (1-x)^{n-k}$

$v' = (1+x)^{n+k}$

$du = -(n-k) \cdot (1-x)^{n-k-1} dx$

$v = \frac{(1+x)^{n+k+1}}{(n+k+1)}$ ✓

Damit gilt:

$$\begin{aligned} \int_{-1}^1 (1-x)^{n-k} \cdot (1+x)^{n+k} dx &= \left[(1-x)^{n-k} \cdot \frac{(1+x)^{n+k+1}}{(n+k+1)} \right]_{-1}^1 + \\ &+ \frac{n-k}{n+k+1} \cdot \int_{-1}^1 (1-x)^{n-k-1} \cdot (1+x)^{n+k+1} dx \end{aligned}$$

✓

Für $x=1$: $(1-1)^{n-k} = 0$

$x=-1$: $(1+(-1))^{n+k+1} = 0$

$\Rightarrow \left[(1-x)^{n-k} \cdot \frac{(1+x)^{n+k+1}}{(n+k+1)} \right]_{-1}^1 = 0 - 0 = 0$ ✓

Da $(n-k)! = (n-k) \cdot (n-k-1)!$ gilt:

$$\frac{(n!)^2}{(n-k)! (n+k)!} \cdot \frac{n-k}{n+k+1} = \frac{(n!)^2}{(n-k-1)! \cdot (n+k+1)!}$$

$$\Rightarrow \int_{-1}^1 (1-x)^n (1+x)^n dx = \frac{(n!)^2}{(n-k-1)! \cdot (n+k+1)!} \cdot \int_{-1}^1 (1-x)^{n-k-1} \cdot (1+x)^{n+k+1} dx$$

□

$$c). \int_{-1}^1 P_n(x)^2 dx = \frac{1}{(2^n n!)^2} \cdot \int_{-1}^1 \frac{d^n}{dx^n} (x^2-1)^n \cdot \frac{d^n}{dx^n} (x^2-1)^n dx$$

Nutzen (a) mit $m=n$, $k=n$:

$$\int_{-1}^1 \frac{d^n}{dx^n} (x^2-1)^n \cdot \frac{d^n}{dx^n} (x^2-1)^n dx = (-1)^n \int_{-1}^1 (x^2-1)^n \cdot \frac{d^{2n}}{dx^{2n}} (x^2-1) dx$$

Polynom $(x^2-1)^n$ hat Grad $2n$ mit 1 als lead. Kf., dann

$$\frac{d^{2n}}{dx^{2n}} (x^2-1)^n = (2n)! \quad \checkmark$$

Nutzen (b) mit $k=n$: $\checkmark$

$$\int_{-1}^1 (1-x)^n (1+x)^n dx = \frac{(n!)^2}{0! \cdot (2n)!} \cdot \int_{-1}^1 (1+x)^{2n} dx =$$

$$= \frac{(n!)^2}{(2n)!} \cdot \int_{-1}^1 (1+x)^{2n} dx$$

$$\int_{-1}^1 (1+x)^{2n} dx = \left[\frac{(1+x)^{2n+1}}{2n+1} \right]_{-1}^1 = \frac{2^{2n+1}}{2n+1}$$

$\checkmark$

Da $(x^2-1)^n = (-1)^n (1-x)^n (1+x)^n$ gilt:

$$\begin{aligned}
 (-1)^n \int_{-1}^1 (x^2 - 1)^n \cdot (2n)! \, dx &= (-1)^n \cdot (2n)! \cdot (-1)^n \cdot \int_{-1}^1 (1-x)^n (1+x)^n \, dx = \\
 &= (2n)! \cdot \frac{(n!)^2}{(2n)!} \cdot \frac{2^{2n+1}}{2n+1} \\
 &= \frac{(n!)^2 \cdot 2^{2n+1}}{2n+1}
 \end{aligned}$$

$$\therefore \int_{-1}^1 P_n(x)^2 \, dx = \frac{1}{(\cancel{2^n n!})^2} \cdot \frac{(\cancel{n!})^2 \cdot \cancel{2^{2n+1}}}{2n+1} = \frac{2^n}{2n+1} \quad \checkmark \quad \square$$

Auf. 1.2.

LHS: $\lim_{n \rightarrow \infty} \int_0^1 \frac{n^2 x}{(1+n^2 x^2)^2} dx$

Substitution: $u = 1 + n^2 x^2$
 $\frac{du}{dx} = 2n^2 x \Rightarrow dx = \frac{du}{2n^2 x}$

$x=0 \Rightarrow u=1$

$x=1 \Rightarrow u = 1 + n^2$
 $\int_1^{1+n^2} \frac{n^2 x}{u^2} \cdot \frac{du}{2n^2 x} = \int_1^{1+n^2} \frac{1}{u^2} \cdot \frac{1}{2} du = \frac{1}{2} \left[-\frac{1}{u} \right]_1^{1+n^2} = \frac{1}{2} \left(-\frac{1}{1+n^2} + \frac{1}{1} \right) =$
 $= \frac{1}{2} \left(\frac{1+n^2-1}{1+n^2} \right) = \frac{n^2}{2(1+n^2)}$

$\lim_{n \rightarrow \infty} \frac{n^2}{2(1+n^2)} = \lim_{n \rightarrow \infty} \frac{1}{\frac{2}{n^2} + 2} = \frac{1}{2} \checkmark$

RHS: $\int_0^1 \lim_{n \rightarrow \infty} \frac{n^2 x}{(1+n^2 x^2)^2} dx \quad x \in [0, 1]$

$x=0$: $\lim_{n \rightarrow \infty} \frac{n^2 \cdot 0}{(1+n^2 \cdot 0)^2} = 0 \checkmark$

$x > 0$: $\lim_{n \rightarrow \infty} \frac{n^2 \cdot x}{(1+n^2 x^2)^2} = \lim_{n \rightarrow \infty} \frac{n^2 x}{1+2n^2 x^2+n^4 x^4} = 0$

$\int_0^1 \lim_{n \rightarrow \infty} f_n(x) dx = \int_0^1 0 dx = 0 \Rightarrow \frac{1}{2} \neq 0 \checkmark$

Widerspruch zu Satz 6.30

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx, \text{ wenn } f_n \text{ konvergiert \& gleichm\"a\ss\!ig!}$$

$$f_n(x) = \frac{\sqrt{27n}}{16} \Rightarrow \lim_{n \rightarrow \infty} \frac{\sqrt{27n}}{16} = \infty \Rightarrow \text{konv. nicht glm.}$$

↳ wegen Norm-Bed. -0,5

Auf. P.3

$$a) \int_0^1 \sqrt{|x \sqrt{x} \sqrt{x}|} dx \quad x \in [0,1] \Rightarrow \int_0^1 \sqrt{x \sqrt{x} \sqrt{x}} dx$$

$$\int_0^1 (x \cdot (x^{\frac{1}{2}})^{\frac{1}{2}})^{\frac{1}{2}} dx \Leftrightarrow \int_0^1 x^{\frac{3}{8}} dx = \left[\frac{8}{15} x^{\frac{15}{8}} \right]_0^1 = \frac{8}{15}$$

$$\Leftrightarrow (x \cdot (x^{\frac{1}{2}})^{\frac{1}{2}})^{\frac{1}{2}} = (x \cdot (x^{\frac{3}{2}})^{\frac{1}{2}})^{\frac{1}{2}} = (x^{\frac{4}{2}} \cdot x^{\frac{3}{4}})^{\frac{1}{2}} = (x^{\frac{7}{4}})^{\frac{1}{2}} = x^{\frac{7}{8}} \quad \checkmark$$

$$b) \int_0^1 e^{x^2} x^3 dx = \int_0^1 e^{x^2} x^2 \cdot x dx \stackrel{x^2=u}{=} \int_0^1 e^u \cdot u \cdot x \cdot \frac{du}{2x} = \frac{1}{2} \int_0^1 e^u \cdot u du$$

$$x=0 \Rightarrow u=0$$

$$x=1 \Rightarrow u=1$$

$$\frac{du}{dx} = 2x \Rightarrow dx = \frac{du}{2x}$$

$$\begin{aligned} \text{P.J.} &= \frac{1}{2} \left[\underbrace{u \cdot e^u}_{=0} - \underbrace{e^u}_{-1} \right]_0^1 = \frac{1}{2} \left[e^u (u-1) \right]_0^1 \\ &= \frac{1}{2} (e^1(1-1) - e^0(0-1)) = \frac{1}{2} \quad \checkmark \end{aligned}$$

$$c) \int_0^{\frac{\pi}{4}} \frac{x}{\cos^2(x)} dx = \int_0^{\frac{\pi}{4}} x \cdot \frac{1}{\cos^2(x)} dx \stackrel{\text{Sub}}{=} \int_0^1 \overbrace{\arctan(u)}^{f(u)} \overbrace{(1+u^2)}^{g'(u)} du =$$

$$u = \tan(x) \quad x=0 \Rightarrow u=0$$

$$\frac{du}{dx} = \frac{1}{\cos^2(x)} \quad x = \frac{\pi}{4} \Rightarrow u=1$$

$$\Rightarrow du = \frac{dx}{\cos^2(x)}$$

$$\stackrel{P.J}{=} \arctan(u) \cdot u \Big|_0^1 - \int_0^1 \frac{1}{1+u^2} \cdot u du$$

$$= \frac{\pi}{4} - 0 - \frac{1}{2} \ln(2) \quad \checkmark$$

$$\int_0^1 \frac{1}{1+u^2} u du = \int_1^2 \frac{1}{z} u \frac{dz}{2u} = \frac{1}{2} \int_1^2 \frac{1}{z} dz = \left[\frac{1}{2} \ln|z| \right]_1^2$$

$$z = 1+u^2$$

$$\frac{dz}{du} = 2u \Rightarrow du = \frac{dz}{2u}$$

$$= \frac{1}{2} \ln(2) - 0 \quad \checkmark$$

$$u=0 \Rightarrow z=1$$

$$u=1 \Rightarrow z=2$$